

Q1. What approach did we take, and what did we reject?

Our approach: Trust Score = A weighted 7-signal composite blending location-bound camera capture, hardware GPS accuracy ($\leq 25\text{m}$), spatial duplicate proximity, and Gemini vision triage, routing scores < 40 to rejection, $40\text{--}70$ to zonal review, and > 70 to priority dispatch.

Citizens capture incidents strictly through an in-app live camera with tamper-resistant GPS coordinates, horizontal accuracy, and device timestamps. Our backend executes a two-stage deduplication filter (50m spatial radius and Levenshtein title matching), pairs it with a Gemini vision model to verify category and hazard severity, and factors in reporter trust reputation. Low-confidence submissions or boundary discrepancies automatically route to the ward sanitary inspector's review queue rather than being dropped.

Alternative we considered and rejected: Mandatory OTP phone authentication and Aadhaar e-KYC before submitting any complaint.

This seemed attractive initially because it guaranteed identity accountability, eliminated automated bot spam, and provided clean audit trails for municipal ward officers.

Q2. Why did we reject it? The trade-off

Dimension	Our approach (Evidence Scoring)	Rejected alternative (Mandatory OTP / KYC)
Citizen Adoption	Instant filing in $< 30\text{s}$ without account hurdles	High drop-off ($> 60\%$) from SMS latency & privacy fears
Offline / 4G Reliability	Local queue stores evidence; syncs when online	Fails in low-connectivity ward edges without SMS
Build Effort in 72h	Focused on spatial indexing, vision, & RBAC	Heavy SMS gateway integration & fallback handling
Spam / Sybil Resistance	Sensor proofs (live camera + GPS + dedup radius)	Strong cryptographic identity verification

Reporting friction and offline reliability at the MCC-panchayat boundary decided our choice; requiring phone verification locks out daily-wage workers, migrants, and citizens in low-signal pockets. The cost we knowingly accepted is that bad actors with real phones can submit false reports that land in our $40\text{--}70$ human-review band, consuming officer triage bandwidth.

Q3. What breaks at the scale of all of Mysuru?

Mysuru comprises 65 municipal wards and surrounding peri-urban gram panchayats, generating 600+ tonnes of solid waste daily, surging to 800+ TPD with 3,500+ daily grievances during the 10-day Dasara festival.

What breaks first	Why (with a rough number)	How we'd fix it
Spatial Duplicate Check	Checking 15,000+ open Dasara items via $O(N)$ distance loops degrades insert throughput ($> 2.4\text{s}$)	Migrate to MongoDB 2dsphere index with geohash-7 cell bucketing ($< 50\text{m}$ bounds)
Zonal Review Queue	3,500 reports/day floods 7 zonal officers with ~ 700 manual audits/day	Auto-cluster contiguous micro-hotspots so officers verify an entire block at once
Vision Model Latency	Burst uploads of 5MB camera photos saturate vision API rate limits	Client-side canvas compression (max 1024px) with asynchronous BullMQ queues

The single change we would make first is replacing brute-force spatial scans with geohash indexing, because insert latency directly stalls offline client sync queues across all 65 wards.